HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2024)

Student No.: 20 Name: _____ Marks: / 23

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. The base of a solid right pyramid is a square of side 64 cm. The height of the pyramid is 24 cm. The pyramid is divided into a frustum X and a pyramid Y by a plane which is parallel to its base. It is given that the height of Y is 18 cm.

   (a) Find the volume of X. (3 marks)

   (b) The base of another solid right pyramid is a square. This pyramid is divided into a frustum Z and a pyramid by a plane which is parallel to its base. The height and the total surface area of Z are 3 cm and 960 cm^2 respectively. Are X and Z similar? Explain your answer. (4 marks)

   (7 marks) (DSE-2024 PI Q13)

Answers written in the margins will not be marked.

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2. Let $f(x) = 2x^2 + 4mx + 8x + 2m^2 + 8m + n$, where m and n are real constants such that $mn < 0$. Denote the vertex of the graph of $y = f(x)$ by P.

(a) Using the method of completing the square, express the coordinates of P in terms of m and n. (2 marks)

(b) Describe the geometric meaning represented by transforming $f(x)$ to $f\left(\frac{x}{5}\right) + 7$. (2 marks)

(c) Denote the vertex of the graph of $y = f\left(\frac{x}{5}\right) + 7$ by Q. Let (a_1, b_1) and (a_2, b_2) be the coordinates of P and Q respectively. It is given that a_1, $1 + n$, a_2 is an arithmetic sequence and b_1, $4 - m$, b_2 is a geometric sequence.

(i) Find the coordinates of P and Q.

(ii) The coordinates of the points R and S are $(3t + 27, t)$ and $(3t + 3, 2t - 3)$ respectively, where t is a real number. Is it possible that $PQRS$ is a rhombus? Explain your answer. (8 marks)

(12 marks) (DSE-2024 PI Q19)

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3. In a school, 40% of the students are girls and $\beta\%$ of the girls are foreign students. It is given that 30% of the boys in the school are foreign students. In the school, the number of foreign students and the number of girls are equal. Find β.

A. 20

B. 45

C. 55

D. 80

(2 marks) (DSE-2024 PII Q11)

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4. $AF000000000BC_{16} =$

A. $175 \times 16^{11} + 188$.

B. $192 \times 16^{11} + 205$.

C. $175 \times 16^{12} + 188$.

D. $192 \times 16^{12} + 205$.

(2 marks) (DSE-2024 PII Q32)

I can finish _____ questions

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